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From Defeasible Logic to Counterfactual Reasoning

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Abstract. Counterfactual reasoning has been the subject of extensive study in philosophy, logics, and AI. The connection between counterfactual reasoning and theory revision is well-known since Ramsey's intuition, according to which "to find out whether the counterfactual 'if A were true, then B would be true' is satisfied in a state S, change the state S minimally to include A, and test whether B is satisfied in the resulting state". In this paper we study how to model this idea in Defeasible Logic for devising logics for counterfactual reasoning and suitable selection function models.

Keywords: Defeasible Logic; Theory revision; Counterfactual reasoning

1 Introduction

What if everyone got the COVID-19 vaccine, would the pandemics be over? What if JFK had survived the assassination attempt? These modes of reasoning have been the subject of extensive study in philosophy, logics, and AI, among others. As is well-known, the first question exemplifies a case of subjective conditional, while the second is an example of counterfactual, i.e., a subjunctive conditional whose antecedent is assumed to be false.

Subjective conditionals and counterfactuals have been notably modeled through two-place modal operators (see [9,11] for an overview). In this paper we study counterfactuals from another, but related, perspective. We are interested in identifying formal relations between counterfactual logics and Defeasible Logic (DL), which is an efficient rule-based non-monotonic formalism where the knowledge base consists of a finite set F of facts, a set of rules, and a priority relation to solve conflicts among rules [2]. Despite the wide variety of applications of DL, to the best of our knowledge, no research has been so far developed in this regard.

Suppose that the following rules govern the dynamic spread of COVID-19 infections:

$$\begin{aligned} R = \{ & r_1 : \mathit{positive}, \mathit{quarantine} \Rightarrow \neg \mathit{spread} \\ & r_2 : \mathit{positive} \Rightarrow \mathit{spread} \\ & r_3 : \mathit{positive}, \mathit{mask} \Rightarrow \neg \mathit{spread} \\ & r_4 : \mathit{spread}, \mathit{vax} \Rightarrow \neg \mathit{high_lethality} \\ & r_5 : \mathit{spread}, \mathit{old} \Rightarrow \mathit{high_lethality} \\ & r_6 : \mathit{high_lethality} \Rightarrow \mathit{hospital_collapse} \} \end{aligned}$$

and assume that r_1 and r_3 are stronger than r_2, while r_4 is stronger than r_5. Also, suppose that the following facts hold:

$$F = \{\mathit{positive}, \mathit{quarantine}, \neg \mathit{mask}, \mathit{old}\}.$$

One can consider different and hypothetical scenarios where

1. quarantine does not work: we could thus change or remove r_1 from R;
2. vax does not prevent high lethality: we could thus make r_4 no longer stronger than r_5;
3. people are not old: we should remove *old* from F.

In general, this intuition suggests that we can model counterfactual reasoning by changing the knowledge base of Defeasible Logic from three different angles. From the theoretical standpoint, there is nothing radically new in this analysis, which is just an elaboration in Defeasible Logic of the original intuition behind Ramsey's test [13]:

> To find out whether the counterfactual "if A were true, then B would be true" is satisfied in a state S, change the state S minimally to include A, and test whether B is satisfied in the resulting state.

However, to the best of our knowledge, no work has proposed minimal change operations over non-monotonic theories that are constructively used to build similarity semantics for counterfactuals.

In this paper we study in Defeasible Logic two mechanisms for changing theories to perform various types of counterfactual reasoning. In particular, we show how to devise semantic structures for counterfactuals that are generated by

1. minimally changing the facts of the knowledge base of Defeasible Logic,
2. contracting or revising the set of defeasible rules of the knowledge base.

In the first case, the resulting sets facts are used to define worlds that are the most similar to the actual one. In the second case, the set of conclusions of the original theory corresponds to the formulas true at the actual world, while the sets of conclusions that follow from the changed sets of rules are used to build an appropriate selection function.

Several issues raise. For example:

- Should the revision of theories be minimal? Would such an idea of minimality be close to the classic concept of similarity between worlds in Lewis' sense [9]?

– What other properties should the facts revision enjoy?

The layout of the paper is as follows. Sections 2 and 3 recall, respectively, the basics of Standard Defeasible Logic and of counterfactual logics (a specific axiomatisation is mentioned simply for the sake of illustration). Section 4 presents a variant of selection function semantics where the function is defined by minimally changing the set of facts of a given Defeasible Theory. Section 5 works instead on contracting or revising the set of rules of a given theory. Some basic results are sketched in Section 6. Brief conclusions end the paper.

2 Defeasible Logic

Let us briefly recall the basics of Standard Defeasible Logic (**SDL**) [2]. We start by defining the language $L_{\mathbf{SDL}}$.

Let PROP be the set of propositional atoms, then the set of literals Lit = PROP $\cup \{\neg p \mid p \in \text{PROP}\}$. The *complementary* of a literal p is denoted by $\sim p$: if p is a positive literal q then $\sim p$ is $\neg q$, if p is a negative literal $\neg q$ then $\sim p$ is q. Literals are denoted by lower-case Roman letters. Let Lab be a set of labels to represent names of rules, which will be denoted as lower-case Greek letters.

A defeasible theory D is a tuple $(\mathrm{F}, R, \prec)$, where F is the set of facts (indisputable statements), R is the rule set, and $\prec$ is a binary relation over R.

R is partitioned into three distinct sets of rules, with different meanings to draw different "types of conclusions". *Strict rules* are rules in the classical fashion: whenever the premises are the case, so is the conclusion. We then have *defeasible rules* which represent the non-monotonic part (along which defeaters) of the logic: if the premises are the case then typically the conclusion holds as well unless we have contrary evidence that opposes and prevents us to draw such a conclusion. Lastly, we have *defeaters*, which are special rules whose purpose is to prevent contrary evidence to be the case. It follows that in DL, through defeasible rules and defeaters, we can represent in a natural way exceptions (and exceptions to exceptions, and so forth).

We finally have the superiority relation $\prec$ a binary relation among couples of rules, that is the mechanism to solve conflicts. Given the two rules α and β, we have $(\alpha, \beta) \in \prec$ (or simply $\alpha \prec \beta$), in the scenario where both rules may fire (can be activated), α's conclusion will be preferred to β's.

A rule $\alpha \in R$ has the form $\alpha\colon A(\alpha) \rightsquigarrow C(\alpha)$, where: (i) $\alpha \in$ Lab is the unique name of the rule, (ii) $A(\alpha) \subseteq$ Lit is α's (set of) antecedents, (iii) $C(\alpha) = l \in$ Lit is its conclusion, and (iv) $\rightsquigarrow \in \{\rightarrow, \Rightarrow, \rightsquigarrow\}$ defines the type of rule, where: $\rightarrow$ is for strict rules, $\Rightarrow$ is for defeasible rules, and $\rightsquigarrow$ is for defeaters.

Some standard abbreviations. The set of strict rules in R is denoted by R_s, and the set of strict and defeasible rules by Rs; $R[l]$ denotes the set of all rules whose conclusion is l.

A *conclusion* of D is a *tagged literal* with one of the following forms:

$\pm\Delta l$ means that l is *definitely proved* (resp. *strictly refuted/non provable*) in D, i.e., there is a definite proof for l in D (resp. a definite proof does not exist).

$\pm\partial l$ means that l is *defeasibly proved* (resp. *defeasibly refuted*) in D, i.e., there is a defeasible proof for l in D (resp. a definite proof does not exist).

The definition of proof is also the standard in DL. Given a defeasible theory D, a proof P of length n in D is a finite sequence $P(1), P(2), \ldots, P(n)$ of tagged formulas of the type $+\Delta l$, $-\Delta l$, $+\partial l$, $-\partial l$, where the proof conditions defined in the rest of this section hold. $P(1..n)$ denotes the first n steps of P.

All proof tags for literals are standard in DL [2]. We report only the positive ones as the negative proof tags can be straightforwardly obtained by applying the *strong negation principle* to the positive counterparts. The definition of Δ describes forward chaining of strict rules.

Definition 1 ($+\Delta$).

$+\Delta l$: *If* $P(n+1) = +\Delta l$ *then either*
(1) $l \in \mathrm{F}$, *or*
(2) $\exists \alpha \in R_s[l]. \forall a \in A(\alpha). + \Delta a \in P(1..n)$.

A literal is strictly proved if it is a(n initial) fact of the theory, or there exists a strict rule that is applicable.

Defeasible derivations are based on the notions of a rule being applicable or discarded. A rule is *applicable* at a given derivation step when every antecedent has been proved at any previous derivation step. Symmetrically, a rule is *discarded* when at least one antecedent has been previously refuted.

Definition 2 (Applicable & Discarded).
Given a defeasible theory D, a literal l, and a prove $P(n)$, we say that

- $\alpha \in R[l]$ *is* applicable *at* $P(n+1)$ *iff* $\forall a \in A(\alpha). + \partial a \in P(1.n)$.
- $\alpha \in R[l]$ *is* discarded *at* $P(n+1)$ *iff* $\exists a \in A(\alpha). - \partial a \in P(1.n)$.

Note that a strict rule can be used to derive defeasible conclusions when is applicable and at least one of its premises is defeasibly but not strictly proved.

Definition 3 ($+\partial$).

$+\partial l$: *If* $P(n+1) = +\partial l$ *then either*
(1) $+\Delta l \in P(1..n)$, *or*
(2.1) $-\Delta{\sim}l \in P(1..n)$, *and*
(2.2) $\exists \alpha \in R[l]$ *applicable s.t.*
(2.3) $\forall \beta \in R[{\sim}l]$ *either*
(2.3.1) β *discarded, or*
(2.3.2) $\exists \zeta \in R[l]$ *applicable s.t.* $\zeta \prec \beta$.

A literal is defeasibly proved if (1) it has already proved as a strict conclusion, or (2.1) the opposite is not and (2.2) there exists an applicable, defeasible or strict, rule such that any counter-attack is either (2.3.1) discarded or (2.3.2) defeated by an applicable, stronger rule supporting l. Note that, whereas β and ζ may be defeaters, α *may not*, as we need a strict or defeasible, applicable rule to draw a conclusion.

The last notions introduced in this section are those of extension of a defeasible theory and of equivalence of theories. Informally, an extension is everything that is derived and disproved; two theories are *equivalent* iff they have the same extension

Definition 4 (Theory Extension). *Given a defeasible theory D, we define the set of positive and negative conclusions of D as its* extension*:*

$$E(D) = (\Delta^+(D), \Delta^-(D), \partial^+(D), \partial^-(D)),$$

where $\#^{\pm}(D) = \{l \mid l$ *appears in* D *and* $D \vdash \pm\#l\}$, $\# \in \{\Delta, \partial\}$.

Theorem 1. *[10] Given a defeasible theory D, its extension $E(D)$ can be computed in time polynomial to the size of the theory. (The size of the theory is given by the number of symbols in it.)*

3 Counterfactuals

Let $L_>$ be the language obtained by adding the conditional connective $>$ to propositional language L. The set of (well-formed) formulas of $L_>$ is defined in the usual way. Formulas of $L_>$ are interpreted in terms of Lewis-type semantic structures [6]. While some, like [1], rejected nested conditionals when a probabilistic treatment of them is considered, others admitted this possibility even though it was argued that the intuitive meaning of such formulas is far from clear [9,11,7]. To keep the logical constructions simple, we follow this view and avoid nested counterfactuals.

The following definitions are standard from [6].

Definition 5. *A* minimal selection function *(MSF)* frame *is a pair $M = \langle W, f\rangle$ where*

1. *W is a nonempty set (of possible worlds);*
2. *f is a selection function $\mathcal{P}(W) \times W \mapsto \mathcal{P}(\mathcal{P}(W))$.*

Definition 6. *A* minimal selection function *(MSF)* model *is a triple $M = \langle W, f, \upsilon\rangle$ where*

1. *W and f are as in Definition 5;*
2. *υ is a valuation assigning to each u in W and $A \in L$ an element from the set $\{T, F\}$.*

In fact, the function f picks out a subset $f(A, u)$ of W for each u in W and $A \in L$.[5] We refer to the set of worlds $f(A, u)$ as the most similar set of worlds with respect to u.

Truth of a formula A at a world u in a model M, $\models_u^M A$, is defined as usual with the counterfactual case given by

$$\models_u^M A > B \text{ iff } \|B\|_M \in f(A, u) \tag{1}$$

[5]Notice that $A \notin L_>$ since we do not admit nested conditionals.

where $\|B\|_M$ denotes the set of worlds where B is true in M, i.e., $\|B\|_M = \{w \in W : \upsilon(B, w) = T\}$: we omit the subscript M when it is clear from the context the model in which a formula is evaluated. A formula A is valid in a model M ($\models^M$) just when $\models^M_u A$ for all worlds u in M. A formula A is valid ($\models^{SF}$) just when $\models^M_u A$ for all worlds u in all *SF* models.

Perhaps one of the weakest systems of counterfactuals requires to adopt the following semantics [12,6].

Definition 7. *A* minimal selection function counterfactual frame *(MSFC) is an MSF frame $M = \langle W, f, \rangle$ satisfying the following conditions:*

1. $X \in f(X, u)$ *(Success)*
2. $\{u\} \in f(X, u)$ *if* $u \in X$ *(Weak Centering)*

Notice that Success ensures that $f(A, u)$ also contains the set of A-worlds. It is not hard to see that the class of *MSFC* frames fits exactly the conditional logic which contains classical propositional logic, the following axioms

1. $A > A$ (**ID**)
2. $(A > B) \rightarrow (A \rightarrow B)$ ($>$ **MP**)

and is closed under the usual inference rules

$$\frac{A \equiv B}{(A > C) \equiv (B > C)} \quad (\mathbf{RCEA})$$

$$\frac{B \equiv C}{(A > B) \equiv (A > C)} \quad (\mathbf{RCEC})$$

As is well-known, the logic consisting of **RCEA** and **RCEC** is the system **CE** [6]. A standard Henkin-style construction proves the completeness of **CE** $\oplus$ **ID** $\oplus >$ **MP** with respect to the class of *MSFC* frames. Proofs are standard and are omitted. Let us call this system **S** (see [6,5]).

Theorem 2. $\models^{SFC} A$ *iff* $\vdash_{\mathbf{S}} A$.

4 Counterfactuals in Defeasible Logic: Fact Revision

Consider a counterfactual conditional

$$\phi > \psi \tag{2}$$

As is well-known, on similarity analysis of genuine counterfactuals we hold in possible-world semantics the following:

> $\phi > \psi$ is true in any world w_0 in case all ϕ-worlds most similar worlds to w_0 are ψ-worlds.

In SDL, recall the COVID example from Section 1 and assume that we want to block the derivation of *hospital_collapse* without imposing a quarantine, thus by removing *quarantine* from F. We can abuse the notation and see the issue as follows: if $w_0 = F$

$$\neg quarantine > \neg hospital_collapse \tag{3}$$

Then we can argue that (3) is true in w_0 iff we derive $\neg$*hospital_collapse* (or, alternatively, we block the derivation of *hospital_collapse*) by revising F at least by removing *quarantine* (or replacing *quarantine* with $\neg$*hospital_collapse*) in such a way as to minimize the effect of such a revision of F.

The following section formally reconstructs the above intuition in a suitable version of selection function semantics.

4.1 Preliminaries

Definition 8 ([4]). *Let $D = (F, R, \prec)$ be any Defeasible Theory. The non-monotonic consequence relation $\mid\!\sim_D$ is defined as follows:*

- $F \mid\!\sim_D l$ *iff* $D \vdash +\partial l$;
- $F \not\mid\!\sim_D l$ *iff* $D \vdash -\partial l$.

Remark 1. Trivially, given any $D = (F, R, \prec)$ and according to Definition 4 the extension $E(D) = (\Delta^+(D), \Delta^-(D), \partial^+(D), \partial^-(D))$ is such that

- $F \mid\!\sim_D l$ iff $l \in \partial^+(D)$
- $F \not\mid\!\sim_D l$ iff $l \in \partial^-(D)$.

Definition 9 (Well-formed propositional formulas). *Let* Lit *be the set of literals of $L_{\mathbf{SDL}}$. The set* WFF *of* well-formed propositional formulas *is defined as follows:*

$$\mathrm{WFF} = \{P | P \in \mathrm{Lit} \cup \{\bigwedge_{i=0}^{n} p | p \in \mathrm{Lit}\}\}.$$

Henceforth, we use the expression "formula" to mean a well-formed propositional formula. A counterfactual is an expression $A > B$ where A, B are well-formed propositional formulas. Thus, as recalled in the previous sections, we do not admit nested conditionals.

Definition 10 (Minimal revision). *Let X be any set of formulas and let $(X)^*_A$ denote the usual AGM-operation of revision. We write $(X)^{*\min}_A$ iff, $\forall Y \supseteq X$: $A \in Y$, $(X)^*_A \subseteq Y$.*

Definition 11. *For any set F of literals, let $(F)^-_q$ be the usual AGM operation of contraction in F of q. Let us stipulate that*

$$F^-_{\{q_1 \wedge \cdots \wedge q_n\}} := F^-_{\{q_1, \ldots, q_n\}} := (\ldots(((F)^-_{q_1})^-_{q_2})\ldots)^-_{q_n}$$
$$F^{*\min}_{\{q_1 \wedge \cdots \wedge q_n\}} := F^{*\min}_{\{q_1, \ldots, q_n\}} := (\ldots(((F)^{*\min}_{q_1})^{*\min}_{q_2})\ldots)^{*\min}_{q_n}$$

The above definition is very useful because of the following proposition.

Proposition 1. *Let $X = l_1, \ldots, l_n$ and $Y = \{q_1, \ldots, q_m\}$ be two consistent sets of literals. For any $q_i, q_k \in Y$, the contraction and minimal revision are commutative, i.e.,*

$$(((X)^-_{q_i})^-_{q_k}) = (((X)^-_{q_k})^-_{q_1})$$
$$(((X)^{*\min}_{q_i})^{*\min}_{q_k}) = (((X)^{*\min}_{q_k})^{*\min}_{q_i})$$

Proof. The proof is trivial, since X and Y are consistent sets of literals. Hence, whatever order of contraction and revision we follow, the operations only affect the literal considered.

Definition 12 (D-p-minimality). *Let $D = (F, R, \prec)$ be any Defeasible Theory. For any literal p, the Defeasible Theory $D' = (F', R, \prec)$ is D-m-p-*critical *iff*

1. *F is consistent;*
2. *$F' \mid\!\sim_{D'} p$, and*
3. *for some consistent set of literals $\{q_1, \ldots, q_m\}$, either*
   (a) *if $\{q_1, \ldots, q_m\} \not\subseteq F$, then $F' - \{q_1, \ldots, q_m\} \not\mid\!\sim_{D'} p$ and $F' = F^{*\min}_{\{q_1,\ldots,q_m\}}$; or*
   (b) *if $\{q_1, \ldots, q_m\} \subseteq F$, then and $F \not\mid\!\sim_D p$ and $F' = F^{-}_{\{q_1,\ldots,q_m\}}$.*

*The Defeasible Theory D' is D-p-*minimal *iff there is no Defeasible Theory D'' such that*

- *D'' is D-j-p-critical, and*
- *$j < m$.*

We also say that the set F' of facts of D' is D-p-minimal if D' is is D-p-minimal.

Example 1 (Running Example). Let us illustrate Definition 12 by resuming the Covid example from Section 1 with some adjustments in the set of facts. The theory $D = (F, R, \prec)$ is as follows:

$$\begin{aligned}
F &= \{positive, vax, \neg mask, old\}\\
R &= \{r_1 : positive, quarantine \Rightarrow \neg spread,\\
&\quad r_2 : positive \Rightarrow spread,\\
&\quad r_3 : positive, mask \Rightarrow \neg spread,\\
&\quad r_4 : spread, vax \Rightarrow \neg high_lethality,\\
&\quad r_5 : spread, old \Rightarrow high_lethality,\\
&\quad r_6 : high_lethality \Rightarrow hospital_collapse\}\\
\prec &= \{\langle r_1, r_2\rangle, \langle r_3, r_2\rangle, \langle r_4, r_5\rangle\}.
\end{aligned}$$

First of all, we trivially observe that F is consistent. The extension $E(D)$ is such that[6]

$$F \mid\!\sim_D \{positive, vax, \neg mask, old, spread, \neg high_lethality\}$$

Suppose $F' = F - \{vax\}$. We fall under condition 3.(b) where $\{q_1, \ldots, q_m\} = \{vax\}$ and $p = high_lethality$: in fact, we note that $F' = F^{-}_{\{vax\}}$, $F' \mid\!\sim_{D'}$ *high_lethality* while $F \not\mid\!\sim_D high_lethality$.

Hence, the theory D is D-1-*high_lethality*-critical. Also, such a theory is trivially D-*high_lethality*-minimal because $m = 1$.

[6] We abuse the notation and write $F \mid\!\sim X$ where X is a set to mean that F non-monotonically imply each element of X.

We are now ready to use classical AGM contraction and revision to build suitable selection-function structures.[7]

4.2 Mapping SDL into Counterfactuals - Part I

For our purposes, we need to express that each Defeasible Theory D corresponds to a special SFC model where just one word makes the facts of D true: such a world—the actual world w_0—is fixed in the model. Also, we have to capture the idea that, for any formula A, the A-worlds selected by f are the most similar to the world w_0 where a counterfactual formula is evaluated.

Definition 13 (D-1-counterfactual models). *Let $D = (F, R, \prec)$ be any Defeasible Theory where $F = \{f_1, \ldots, f_n\}$. A* selection function D-1-counterfactual model *(D-1-SFDC model) is a structure $M = \langle W, w_0, f, \upsilon\rangle$ where:*

1. *$w_0 \in W$ is the actual world of M such that $\bigcap_{i=0}^{n} \|f_i\| = \{w_0\}$;*
2. *for any set of literals $\{q_1, \ldots, q_m\}$, $f(\bigwedge_{i=0}^{m} q_i, w_0)$ is a set X of sets of worlds such that, for each $Y \in X$, there is a set of literals $F^w_{f(\bigwedge_{i=0}^{m} q_i, w_0)} = \{p_1, \ldots, p_k\}$ where $Y = \bigcap_{i=0}^{k} \|p_i\|$ and, for some set Z of literals, either*
   - *$F^w_{f(\bigwedge_{i=0}^{m} q_i, w_0)} = F^{*\min}_Z$, or*
   - *$F^w_{f(\bigwedge_{i=0}^{m} q_i, w_0)} = F^-_Z$.*

Definition 14 (Counterfactual 1-evaluation). *Let $D = (F, R, \prec)$ be any Defeasible Theory. The truth of any counterfactual $A > B$ at a world w_0 in an D-1-SFDC model M is defined as follows:*

$$\models^M_{w_0} A > B \tag{4}$$

iff

- *$\|B\| \in f(A, w_0)$;*
- *$F^w_{f(A,w_0)}$ is D-p-minimal.*

The frame conditions of Success and Weak Centering recalled in Definition 7 can be easily adapted for D-1-*SFDC* models.

The language $L_{D>}$ for the counterfactual logic generated by any Defeasible Theory is clearly a fragment of $L_>$.

Let us illustrate the above definitions with our running example and build a toy model.

Example 2 (Running Example (cont'd)). Resume the theory D from Example 1 and build an appropriate D-1-*SFDC* model considering the following literals

$$positive, vax, \neg mask, old, spread, \neg high_lethality, hospital_collapse, quarantine$$

and only three worlds: w_0, w_1, and w_2. Let proceed as follows:

[7] In Section 5 we will go beyond standard AGM operations and exploit specific revision operations devised to handle non-monotonic theories.

– Evaluations of propositional formulas are as follows:

$$\|positive\| = \{w_0, w_1\}$$
$$\|vax\| = \{w_0, w_1, w_2\}$$
$$\|\neg mask\| = \{w_0, w_1\}$$
$$\|old\| = \{w_0, w_1\}$$
$$\|spread\| = \{w_1, w_2\}$$
$$\|\neg high_lethality\| = \{w_1\}$$
$$\|hospital_collapse\| = \{w_1\}$$
$$\|quarantine\| = \{w_1, w_2\}$$

– Clearly, $\|positive\| \cap \|vax\| \cap \|\neg mask\| \cap \|old\| = \{w_0\}$;
– Define, among others,

$$f(\neg vax, w_0) = F^w_{f(\neg vax, w_0)} = \{\{w_1\}\}.$$

where $Z = F - \{vax\} = F^-_{vax}$.

It is easy to check that $F^w_{f(\neg vax, w_0)}$ is D-*high_lethality*-minimal, thus

$$\models^M_{w_0} \neg vax > high_lethality.$$

5 Counterfactuals in Defeasible Logic: Theory Revision

[3] reframed AGM operations for SDL, i.e., investigated revision and contraction operators for defeasible theories, and proposed postulates motivated by the form or the intuition of the AGM postulates for classical belief revision. This result is interesting because we can build counterfactual models by exploiting minimality in AGM sense.

5.1 Revising Defeasible Theories

Let us recall the basic definitions and results of [3]. Let $D = (F, R, \prec)$ be a Defeasible Theory and $c = p_1, \ldots p_n$ be the formulas to considered. By convention, for any literal l, by $l \in E(D)$ we mean that $F \mid\!\sim_D l$.

Definition 15 (Expansion). *Expansion is defined as follows:*

$$D^+_c = \begin{cases} D & \text{if } \sim p_i \in \partial^+(D) \text{ or } \sim p_i = p_j \text{ for some } i, j \in \{1, \ldots, n\} \\ (F, R', \prec') & \text{otherwise} \end{cases}$$

where

$$\begin{aligned} R' =& R \cup \{\Rightarrow p_1, \ldots, \Rightarrow p_n\} \\ \prec' =& (\prec \cup \{\Rightarrow p_i \prec r \mid i \in \{1, \ldots, n\}, r \in R[\sim p_i]\}) - \\ & \{r \prec \Rightarrow p_i \mid i \in \{1, \ldots, n\}, r \in R[\sim p_i]\}. \end{aligned} \tag{5}$$

AGM *revision* works in the same way as AGM expansion when the formula to be added does not cause an inconsistency, but revision adds a formula even if its negation is in the belief set. In [3]'s framework the definition of revision is as follows.

Definition 16 (Revision). *Revision is defined as follows:*

$$D_c^* = \begin{cases} D & \textit{if } p_1, \ldots, p_n \in E(D) \\ (F, R', \prec') & \textit{otherwise} \end{cases}$$
$$\begin{aligned} &\textit{where} \\ &R' = R \cup \{\Rightarrow p_1, \ldots, \Rightarrow p_n\} \\ &\prec' = (\prec \cup \{\Rightarrow p_i \prec r \mid i \in \{1, \ldots, n\}, r \in R[\sim p_i]\}) - \\ &\qquad \{r \prec \Rightarrow p_i \mid i \in \{1, \ldots, n\}, r \in R[\sim p_i]\}. \end{aligned} \tag{6}$$

Definition 17 (Contraction). *Contraction is defined as follows:*

$$D_c^- = \begin{cases} D & \textit{if } p_1, \ldots, p_n \notin \partial^+(D) \\ (F, R', \prec') & \textit{otherwise} \end{cases}$$
$$\begin{aligned} &\textit{where} \\ &R' = R \cup \{p_1, \ldots, p_{i-1}, p_{i+1}, \ldots, p_n \rightsquigarrow \ \sim p_i \mid i \in \{1, \ldots, n\}\} \\ &\prec' = \ \prec - \{s \prec r \mid r \in R' - R, s \in R\}. \end{aligned} \tag{7}$$

Theorem 3. *Operations in Definitions 15, 16, and 17 satisfy the reformulation of AGM postulates given in [3].*

As regards minimality in the AGM sense, we have to define belief sets for SDL.

Definition 18. *The* belief set $B(D)$ *of a Defeasible Theory* D *is defined as follows:*

$$B(D) = \{P | P \in E(D) \cup \{\bigwedge_{i=0}^{n} p | p \in E(D)\} \textit{ and } B(D) \textit{ is closed under sub-formulas}\}$$

Then, it should be noted that the following postulate, among others, holds for any formula c [3]:

$$\text{If } c \notin B(D) \text{ then } B(D)_c^- = B(D). \tag{-3}$$

In general, minimality for modeling counterfactuals is guaranteed by Definitions 16 and 17 since the operations just add the rules only needed, respectively, to prove or to disprove the considered formula. Just notice that SDL is skeptical, so it may happen that revision may lead, e.g., to block other literals that were in the extension, while contraction may contribute to prove new conclusions.

Consider again our running example.

Example 3 (Running Example (cont'd)). In Example 1 we considered how to revise the set of facts by removing *vax* from F. In Example 2, we used this change for building an appropriare semantic structure for counterfactuals. Here operate at a different level, i.e., by considering the defeasible extension of the theory. Suppose we retain F' and aim at applying Definition 17 to contract *spread*:

$$D'' = D^{-}_{spread}(F', R', \prec')$$

where

$$R' = R \cup \{ \rightsquigarrow \sim spread\}$$
$$\prec' = \prec - \{s \prec r \mid r \in R' - R, s \in R\}.$$

It is easy to check that $F' \not\sim_{D''} \{spread, high_lethality, hospital_collapse\}$.

5.2 Mapping SDL into Counterfactuals - Part II

Once minimal contraction and revision for SDL are defined, we can use the extensions of theories and the corresponding belief sets to generate suitable selection function counterfactual models for the logic **S**.

Definition 19 (D-2-counterfactual models). *Let $D = (F, R, \prec)$ be any Defeasible Theory and $E(D) = \{f_1, \ldots, f_n\}$ be its extension. A* selection function D-2-counterfactual model *(D-2-SFDC model) is a structure $M = \langle W, w_0, f, \upsilon\rangle$ where:*

1. *$w_0 \in W$ is the actual world of M such that $\bigcap_{i=0}^{n} \|f_i\| = \{w_0\}$;*
2. *for any set of literals $\{q_1, \ldots, q_m\}$, $f(\bigwedge_{i=0}^{m} q_i, w_0)$ is a set X of worlds such that, for each $w \in X$, there is a set of literals $F^{w}_{f(\bigwedge_{i=0}^{m} q_i, w_0)} = \{p_1, \ldots, p_k\}$ where $w \in \bigcap_{i=0}^{k} \|p_i\|$ and, for some set Z of literals, either*
   - *$F^{w}_{f(\bigwedge_{i=0}^{m} q_i, w_0)} = B(D^{*}_{Z})$, or*
   - *$F^{w}_{f(\bigwedge_{i=0}^{m} q_i, w_0)} = B(D^{-}_{Z})$.*

Using theory extensions to define the set of worlds allows for keeping standard the notion of evaluation:

Definition 20 (Counterfactual 2-evaluation). *Let $D = (F, R, \prec)$ be any Defeasible Theory. The truth of any counterfactual $A > B$ at a world w_0 in an D-2-SFDC model M is defined as follows:*

$$\models^{M}_{w_0} A > B \text{ iff } \|B\| \in f(A, w_0). \tag{8}$$

Example 4 (Running Example (cont'd)). Let us proceed similarly as we did in Example 2. Consider the usual three worlds: w_0, w_1, and w_2. We modify the scenario as follows:

– Evaluations of propositional formulas:

$$\|positive\| = \{w_0, w_1\}$$
$$\|vax\| = \{w_0, w_1, w_2\}$$
$$\|\neg mask\| = \{w_0, w_1\}$$
$$\|old\| = \{w_0, w_1\}$$
$$\|\neg spread\| = \{w_1, w_2\}$$
$$\|\neg high_lethality\| = \{w_1\}$$
$$\|\neg hospital_collapse\| = \{w_1\}$$
$$\|quarantine\| = \{w_1, w_2\}$$

– Clearly, $\|positive\| \cap \|vax\| \cap \|\neg mask\| \cap \|old\| = \{w_0\}$;
– Apply Definition 17 and define, among others (remember that $D'' = D^{-}_{spread}(F', R', \prec')$),

$$f(\neg spread, w_0) = F^{w}_{f(\neg spread, w_0)} = \{\{w_1\}\}.$$

where $Z = B(D'')$.

It is easy to check that the resulting structure M is a D-2-counterfactual model where

$$\models^{M}_{w_0} \neg spread > \neg hospital_collapse.$$

6 Results

Under the language limitations mentioned above for **S**, completeness is a rather easy result for both versions of the semantics and trivial adjustments from [6] can be settled. Indeed, except for the fact that we fix an actual world, selection function D-1 and D-2-counterfactual models are standard.

Hence, let us consider, for the sake of illustration, canonical frames for selection function D-2-counterfactual semantics [6].

Definition 21 (Canonical structures). *Let W be the set of $L_{D>}$-maximal consistent sets of formulas of $L_{D>}$, and f be a selection function $\mathcal{P}(W) \times W \mapsto \mathcal{P}(\mathcal{P}(W))$ where, for any $w \in W$*

$$A > B \in w_0 \text{ iff } |B|_{L_{D>}} \in f(|A|_{L_{D>}}, w_0).$$

The structure $F = \langle W, w_0, f \rangle$ is a canonical frame for $L_{D>}$.

A second, and perhaps less trivial result is that the models generated from Defeasible Theories are related with the notion of filtration [6,5].

Again, let us focus on the second variant of semantics and define the notion of filtration with a fixed actual world.

Definition 22 (Filtrations). *Let $M = \langle W, w_0, f, \upsilon \rangle$ be a D-2-SFDC model, Γ be a finite set of formulas closed under sub-formulas, and $\sim$ be the equivalence relation on the set W of worlds such that*

$$u \sim w \text{ iff, for every } A \in \Gamma, \models^{M}_{u} A \text{ iff } \models^{M}_{w} A$$

$[w]$ is the $\sim$-equivalence class of worlds in M generated by w and $[X] = \{[w] | w \in X \subseteq W\}$.

A filtration of M through Γ is any model $M^\Gamma = \langle W^\Gamma, w_0^\Gamma, f^\Gamma, \upsilon^\Gamma \rangle$ where

- *$W^\Gamma = [W]$;*
- *$w_0^\Gamma = [w_0]$;*
- *f^Γ is a function $\mathcal{P}(W^\Gamma) \times W^\Gamma \mapsto \mathcal{P}(\mathcal{P}(W^\Gamma))$ such that*
  - *$[f(X, w)] = f^\Gamma([X], [w])$;*
  - *$f^\Gamma([X], [w_0]) = [Y]$ where, for every counterfactual $A > B \in \Gamma$ we have that $X = \|A\|_M$, $Y = \|B\|_M$, and $\models^M_{w_0} A > B$.*
- *υ^Γ is a valuation assigning to each u^Γ in W^Γ and $A \in \Gamma$ an element from the set $\{T, F\}$.*

All results proved in [6,5] apply as well to this case. We should note however that an additional property holds. Such a property illustrates under what simple conditions a D-2-counterfactual model generated by a theory is a filtration of some other model.

Theorem 4. *Let $D = (F, R, \prec)$ be any Defeasible Theory, $E(D) = \{f_1, \ldots, f_n\}$ be its extension, and M be a selection function D-2-counterfactual model $M = \langle W, w_0, f, \upsilon \rangle$ where*

1. *w_0 is as in Definition 19;*
2. *for any set of literals $\{q_1, \ldots, q_m\}$, $f(\bigwedge_{i=0}^m q_i, w_0)$ is such that there is no $\{p_1, \ldots, p_n\}$ where $F^w_{f(\bigwedge_{i=0}^n p_i, w_0)} = F^w_{f(\bigwedge_{i=0}^m q_i, w_0)}$.*
   If X_D is the set of literals occurring in D and each $\{q_1, \ldots, q_m\} \subseteq X_D$, then M is a filtration of some model M' through $\bigcup_{j=0}^s Y_j$ such that $\{Y_1, \ldots, Y_s\}$ is the codomain of f.

Proof. Since X_D is finite, f is finite as well, so is $\{Y_1, \ldots, Y_s\}$. By construction w_0 is $[\bigwedge_{i=1}^n f_i]$ while each set of worlds Y_k in $\{Y_1, \ldots, Y_s\}$ is the set of worlds that make true the elements of the corresponding belief set.

7 Conclusions

In this paper we have investigated in Defeasible Logic two mechanisms for changing theories which can be used to devise semantic structures for counterfactual logics.

We have proposed semantic structures for counterfactuals that are generated by

1. minimally changing the facts of the knowledge base of Defeasible Logic,
2. contracting or revising the set of defeasible rules of the knowledge base.

This paper offers a preliminary but fresh idea: to the best of our knowledge, no work has proposed minimal change operations over non-monotonic theories that are constructively used to build similarity semantics for counterfactuals.

Another work has investigated the relation between Defeasible Logic and possible world semantics [8]. However, that work was meant to address a different research challenge, i.e., to develop neighbourhood semantics for a modal

extension of SDL, thus going beyond operational and argumentation semantics for it.

This paper has a rather different goal: interpret Ramsey's test by showing how the change of defeasible theories allows for constructing selection function models for counterfactuals.

One limit of our work is that we exclude nested conditionals from the language. While some, like [1], rejected nested conditionals when a probabilistic treatment is considered, or it was argued that the intuitive meaning of such formulas is not clear [9,11,7], this is still a technical possibility. How to extend our methodology to deal with this case is a matter of future work.

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